Enumerative Combinatorics — problem 5.27

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Euler's ϕ-function of a positive integer n is defined as follows :

$$\phi(n) = \text{the number of positive integers} \leq n \text{ which are prime to } n.$$

Prove that

$$\phi(n) = n \prod_{i=1}^{m} \left(1 - \frac{1}{p_i}\right),$$

where $p_1, p_2, \ldots, p_m$ are all distinct prime divisors of n. Use PIE.